

2021 Ph H2 Q4

Section: Our Dynamic Universe

Topic: The Expanding Universe

The diagram claims to illustrate the evolution of the Universe from the Big Bang to the present, showing various particles and 'eras'. The task is to comment on its accuracy using physics knowledge.

Worked solution

- The diagram is misleading as it mixes fundamental particles (quarks, electrons) with composite particles (protons, neutrons) without a clear timescale.
- It suggests all types of particles coexisted simultaneously, but in reality different eras dominated: the quark era, followed by the formation of hadrons (protons, neutrons), then nuclei (nuclear era), then atoms (atomic era).
- Neutrinos and photons decoupled at specific times — they should not be shown as simply coexisting from the start.
- The diagram gives no indication of the expansion of space or the cooling of the Universe, which are central to Big Bang cosmology.

Final points

The diagram is oversimplified and inaccurate. The proper sequence is: quark-gluon plasma → hadrons (protons, neutrons) → nuclei (Big Bang nucleosynthesis) → atoms (recombination) → galaxies and stars. It should show expansion, cooling, and a timeline.

Revision tips

- In the early Universe: quarks → hadrons → nuclei → atoms → galaxies.
- Cooling and expansion are the key processes driving these changes.
- Beware of oversimplified diagrams: they may mix up particle types or eras.
- Big Bang nucleosynthesis (≈ 3 min after Big Bang) formed light nuclei.
- Recombination ($\approx 380,000$ years later) allowed atoms to form.